4. Forecasting and Time Series Analysis

Intro

Time series analysis and forecasting are crucial for predicting future trends, behaviours based on historical data. It helps businesses make informed decisions, optimize resources and mitigate risks by anticipating market demand, sales fluctuations, stock prices & more.

* **Time Series →**

A time series is a sequence of data points collected, recorded or measured at successive or regular and equally spaced time intervals.

Features →
- ordered in time (chronological order)
- Time is critical aspect of the data.
- Each data point corresponds to time stamp.
- can be daily, weekly, monthly, yearly etc.

e.g. → Daily temperature readings, yearly rainfall stats etc.

* Time series data is often plotted with time on the x-axis and the variable of interest on the y-axis to visually analyze trends, patterns & fluctuations.

* **Forecasting →**

Forecasting is the process of making predictions about future events based on historical events or data and analysis. It plays a vital role in decision-making by providing insight into expected future trends & patterns. It is also estimation of future values using past & present data trends, often through mathematical models & statistical

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* Types →

A] Qualitative Forecasting →

- Based on expert opinions, intuition or market research.
- Used when historical data is unavailable or insufficient.

e.g. Market surveys, expert panels

B] Quantitative Forecasting →

- Relies on historical numerical data & statistical models.
- Most commonly used in time series forecasting.

Methods → Time series Analysis (ARIMA), Regression Analysis ML models etc.

* Steps in Forecasting Process →

1. Problem Definition - Define what needs to be forecasted.

2. Data Collection - Gather relevant historical data.

3. Data Preprocessing - Handle missing values, remove outliers and ensure stationarity.

4. Model Selection - ~~Fit the model on historical data.~~ Choose appropiate forecasting model.

5. Model Training - Fit the model on historical data.

6. Forecast Generation - Use the model to predict future values

7. Model Evaluation - Evaluate model accuracy using metrices like RMSE, MSE or MAPE.

8. Decision Making - Interpret and use forecasted values for strategic planning.

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* Time Series Analysis Vs. Forecasting

	Time Series Analysis	Forecasting
1)	Study of time-ordered to understand underlying patterns.	1) Process of predicting future values based on past data.
2)	Analyze historical behaviour and identify components	2) Estimate or forecast future trends, values or outcomes.
3)	It has focus on understanding trends, seasonality, cycle & noise.	3) It has focus on making accurate future predictions.
4)	It is descriptive & diagonstic.	4) It is predictive & strategic.
5)	It requires historical time series data.	5) It requires historical data used to forecast future values
6)	Here output is insight to the data behaviour (i.e trend, seasonality)	6) Here output is forecasted values or future data points.
7)	It is used for Detecting anomalies, identifying cycles etc.	7) It is used for planning budgets, demand forecasting, stock prediction etc.
8)	Techniques used are ACF/PACF plots, decomposition & Stationarity tests.	8) Techniques used are ARIMA, SARIMA, prophet, LSTM etc.
9)	It is Independent Process.	9) Depend on time series analysis for accurate modeling.

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* Time Series Analysis Components →

A time series is a sequence of data points recorded over regular time intervals. Understanding the components of a time series is essential for effective analysis and forecasting.

Component are:

1) Trend →

The trend component represents the long-term progression or movement in the data. It indicates whether the values are increasing, decreasing or remaining stable over time.

* characteristics →
- can be linear or non-linear.
- Does not repeat in short intervals.
- Usually visible over a long period.

* Types →
1) Uptrend → Time series Analysis shows a general pattern that is upward then it is trend. (up).
2) downtrend → Time series Analysis shows a pattern that is downward then it is down trend.
3) Horizontal or stationary trend → If no pattern observed then it is called a Horizontal or stationary trend.

e.g. → Increasing global temperatures over decades.

2) Seasonality →

Seasonality refers to repeating patterns or fluctuations that occur at regular time intervals due to seasonal factors like weather, holidays or business cycles.

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* Characteristics →
- Predictable and occurs at fixed periods (e.g. daily, weekly)
- Influenced by cultural, social or environmental factors

e.g. Ice cream sales increase in summer every year.

3) Cyclic variations →

Cyclic components are long-term oscillations in a time series that occur in non-fixed intervals, often influenced by economic or business cycles.

* characteristics →
- Unlike seasonality, cycles do not have a fixed frequency.
- Last for more than one year & are influenced by macroeconomic factors.

e.g. Recession and boom phases in an economy.

4) Irregular or Random Component →

This component represents random variations or unpredictable fluctuations in the data that cannot be explained by trend, seasonality or cyclic patterns.

* characteristics →
- caused by unforeseen or unusual events
- Difficult or impossible to model
- Also called noise in data

e.g. Sudden market crash due to a geopolitical crisis.

* Types of Data in Time-Dependent Analysis →

1) Time series Data →

Time series data refers to data collected over successive and evenly spaced time intervals, such as daily,

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month, or yearly. This type of data captures how a particular variable evolves over time.
- The main objective of time series is to analyze trends, seasonal variations and cyclical patterns. It helps in understanding past behaviours and forecasting future outcomes.

e.g. →
- Daily stock market prices
- Monthly electricity consumption

2] Cross-sectional Data →
- Cross-sectional data involves data collected at a single point in time across different subjects or entities. It does not consider the time dimension but focuses on comparing multiple observations simultaneously.
- It is mainly used to analyze relationships or difference between different individuals, groups or entities at a specific time.

e.g. →
- Exam scores of students from different schools in the same year
- Sales figures of multiple products in one month.

3] Pooled Data (Panel Data) →
- Pooled data, also known as panel data, is combination of time series and cross-sectional data. It observes multiple subjects over a period of time.
- This hybrid format allows for more comprehensive and nuanced analysis by capturing both individual-specific effects and temporal dynamics. It is often used in econometrics and social sciences.

e.g. → Annual income data of the same 100 households over 10 years.

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* Stationarity and non-stationarity →

A) Stationarity →

A time series is said to be stationary if its statistical properties such as mean, variance and autocorrelation remain constant over time.

* Characterstics →
- Constant mean over time.
- Constant variance over time.
- Autocorrelation values do not depend on time.
- No trend or seasonal effects.

* It is used in series models like ARMA, ARIMA, etc require the data to be stationary to provide accurate forecasts.

- e.g. → Daily temperature deviations from the average in a climate-controlled room.

B) Non-Stationary Data →

A time series is non-stationary if its mean, variance or autocorrelation change over time.

* Characterstics:
- Presence of trends (increasing & decreasing values)
- Presence of seasonality or periodic fluctuations.
- Varying variance over time (heteroscedasticity)

* Common Transformation to Make Data Stationary:
- Differencing (subtracting current and previous values)
- Log transformation

e.g.
- Monthly sales data showing a steady upward trend.
- Population growth over years.

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* Moving Averages Methodology →

The Moving Average (MA) is a widely used technique in time series analysis to smooth out short-term fluctuations and highlight long-term trends or cycles in the data.

Def^n →

A moving average is a statistical method used to calculate the average of a time series over a specific number of past observations, which moves forward with each new data point.

* Purpose ⇒

- Smoothnens the data to reduce noise or volatility.
- Highlights trends and patterns by removing short-term irregular
- Acts as a basis for forecasting future values in time series.

* Types of Moving Averages →

A] Simple Moving Average (SMA) →

- A simple moving average (SMA) is a technique used in time series analysis to smooth out short-term fluctuations and highlight longer-term trends or cycles.
- It is calculated by taking the arithmetic mean of a fixed number of consecutive data points.

formula → $$SMA_t = \frac{X_t + X_{t-1} + \dots + X_{t-(n-1)}}{n}$$

where;

- SMA_t : Simple Moving Average at time t
- X_t : Value of the time series at time t
- n : Number of time periods

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Explanations →

- The SMA "moves" because as each new data point become available, the oldest data point is dropped, and the newest is added.
- This creates a rolling average, giving a smoothed version of the original time series.

e.g. →

Dataset → 100, 120, 130, 150, 170 (weeks)

→ take window size = 3

1) For Week 3:

$$SMA_3 = \frac{100 + 120 + 130}{3} = \frac{350}{3} = \underline{116.67}$$

2) For week 4:

$$SMA_4 = \frac{120 + 130 + 150}{3} = \frac{400}{3} = 133.33$$

3) For week 5:

$$SMA_5 = \frac{130 + 150 + 170}{3} = \frac{450}{3} = 150$$

↳ window size

Week	Sales	SMA (week-3)
1	100	-
2	120	-
3	130	116.67
4	150	133.33
5	170	150.00

∴ Here there is a upward trend in the moving averages which indicates increasing sales over the weeks.

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$$MAD = \frac{1}{n}\sum_{i=1}^{n} |Actual_i - Forecast_i|$$

↳ Minimum value of MAD means good forecast

B] Weighted Moving Averages →

The weighted moving average (WMA) is a time series smoothing technique that, unlike the SMA, assigns different weights to different observations, usually giving more importance to recent data. This makes WMA more responsive to recent trends in the data.

* Why to use WMA?

while SMA treats all observations equally, WMA recognizes the recent data may be more relevant for predicting future values. This is particularly important in:

- Financial markets
- Weather & climate prediction
- Inventory & Sales management

* Characterstics →

- It finds more recent observations
- Faster reaction to trend shifts compared to SMA.
- It is Customizable means we can adjust weights on specific business models.

* Formula →

$$WMA = \frac{P_1 \times W_1 + P_2 \times W_2 + \ldots\ldots + P_n \times W_n}{W_1 + W_2 + \ldots + W_n}$$

$$WMA_t = \frac{\sum_{i=0}^{n-1} w_i \cdot P_{t-i}}{\sum_{i=0}^{n-1} w_i}$$

where, P → It is value for each day in period

P_{t-i} → Observation at diff. time points

W → Weight assign to each observation.

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e.g → Forecast sales using 4-weeks weighted moving averages with weights 0.4, 0.3, 0.2 & 0.1

week	Sales	4-week WMA	Error
1	39		
2	44		
3	40		
4	45		
5	38	42.7	4.7
6	43	41.1	1.9
7	39	41.6	2.6
8		(40.6)	

XW above side means we have to consider most recent first

Formula → $WMA = \frac{P_1 \times W_1 + P_2 \times W_2 + \dots + P_n W_n}{W_1 + W_2 + \dots + W_n}$

∴ we have to calculate forecast for

1) week 5 →

$$F_5 = \frac{0.4 \times 45 + 0.3 \times 40 + 0.2 \times 44 + 0.1 \times 39}{0.4 + 0.3 + 0.2 + 0.1}$$

∴ WMA_5 or $F_5 = 42.7$ (write at 5 week row in WMA)

$$\text{Error} = WMA_5 - \text{Sales}_5 = |42.7 - 38| = 4.7$$

2) week 6 →

$$F_6 = \frac{0.4 \times 38 + 0.3 \times 45 + 0.2 \times 40 + 0.1 \times 44}{1}$$

$$F_6 = 41.1$$

$$\text{Error} = |41.1 - 43| = 1.9$$

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3) For week 7 →

$$F_7 = \frac{43 \times 0.4 + 38 \times 0.3 + 45 \times 0.2 + 40 \times 0.1}{1}$$

$$F_7 = 41.6$$

Error = $41.6 - 39 = 2.6$

4) For week 8 → (Final Forecast)

$$F_8 = \frac{39 \times 0.4 + 43 \times 0.3 + 38 \times 0.2 + 45 \times 0.1}{1}$$

$$F_8 = 40.6$$

Now, we have to calculate the MAD

Mean absolute deviation (MAD) $= \frac{4.7 + 1.9 + 2.6}{3} = 3.07$

If we calculate WMA for 2-week →

It gives value for $F_8 = 40.6$

but the

MAD = 2.92

So, by comparing the both we can say that the,

2-week's forecast is better than 4-weeks forecast.

Since,

MAD(2W) < MAD (4-W)

∴ $2.92 < 3.07$

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4) Cummulative Moving Average (CMA) →

- The cummulative Moving Average (CMA) is a time series smoothing technique used to track the average of a variable over time.
- Unlike other moving averages that uses a fixed number of recent data points, CMA calculates the average from the begining of the dataset to the current point, continuosly incorporating each new observation.

* Properties :

1) Cummulative Nature : Includes all past data points in the average, making it a comprehensive measure of the overall trend
2) Smooth output : The more data points added, the smoother the average becomes, as random fluctuations get averaged out
3) Easy to update : Require only previous CMA & the current value to compute the next, make it computionally efficient.
4) Lagging Nature : As It includes all data, it reacts slowly to recent changes compared to SMA or EMA.

* Why to use CMA ?

- Long-term Trend Tracking : Captures the overall direction of data by averaging all past values.
- Smoothens Fluctuations : Reduces short-term noise for clearer insights into patterns.
- Simple & efficient : Easy to update using a recursive formula ideal for real-time systems
- Bettor Decision Making : Helps monitor performance & detect gradual Changes over time.

* Formula :

$$CMA_t = \frac{\sum_{i=1}^{t} x_i}{t}$$

← No use in exam

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where,
- CMA_t → CMA at time t
- x_i → data point at time i
- t → No. of data points up to current time

* CMA can be updated iteratively, ← use in exam

$$CMA_t = CMA_{t-1} + \frac{x_t - CMA_{t-1}}{t}$$

e.g → Data → 10, 20, 30, 40, 50, 25, 35

Data	CMA
10	10
20	15
30	20
40	25
50	30
25	29.17
35	30

i] For 10 →

$$CMA_1 = CMA_0 + \frac{x_1 - CMA_0}{t} = 0 + \frac{10}{1} = \underline{\underline{10}}$$

ii] For 20 →

$$CMA_2 = CMA_1 + \frac{x_2 - CMA_1}{t} = 10 + \frac{20-10}{2} = \underline{\underline{15}}$$

iii] For 30 →

$$CMA_3 = CMA_2 + \frac{x_3 - CMA_3}{t} = 15 + \frac{30-15}{3} = \underline{\underline{20}}$$

iv] For 40 →

$$CMA_4 = CMA_3 + \frac{x_4 - CMA_3}{t} = 20 + \frac{40-20}{4} = \underline{\underline{25}}$$

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v] For 50 →

$$CMA_5 = CMA_4 + \frac{x_5 - CMA_4}{t} = 25 + \frac{50 - 25}{1} = 50$$

vi] For 25 →

$$CMA_6 = CMA_5 + \frac{x_6 - CMA_5}{t} = 30 + \frac{25 - 30}{6} = 29.17$$

vii] For 35 →

$$CMA_7 = CMA_6 + \frac{x_7 - CMA_6}{t} = 29.17 + \frac{35 - 29.17}{7} = 30$$

* Disadvantages:

- Sensitive to older data, which may not represent recent trends well
- Not suitable for rapidly changing processes where recent data is more relevant.

D] Exponential Moving Average (EMA)

- The exponential Moving Average (EMA) is a type of weighted moving average used in time series analysis to smooth out data and highlight trends.
- Unlike the simple moving Average (SMA), which gives equal weight to all values in a window, the EMA gives more importance to recent data, making it more responsive to new information.

Defn →

The EMA is calculated by applying exponentially decreasing weights to past observations. It helps in detecting short-term trends more effectively than SMA.

* Features →

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- More responsive to recent data: Recent observations carry more weight.
- Reduces lag: Compared to SMA, EMA reacts faster to changes in the data.
- Recursive Nature: EMA depends on its previous value, making it efficient to compute.

* Steps for calculating EMA →

step 1: Identify Given data

- Time series data: $x_1, x_2, x_3, \ldots, x_n$
- chose periods: N

Step 2: calculate the Multiplier (smoothing factor)

$$\alpha = \frac{2}{N+1}$$

Step 3: Compute SMA of first N values as First EMA:

$$EMA_N = \frac{x_1 + x_2 + x_3 + \ldots + x_N}{N}$$

Step 4: Apply the EMA formula Recursively

$$F_{t+1} = F_t + \alpha (A_t - F_t) \qquad 0 \leq \alpha \leq 1$$

$$\therefore \boxed{F_{t+1} = \alpha A_t + (1-\alpha) F_t}$$

∵ F_t is previous forec

e.g →

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Week	A_t Sales	F_t Forecast
1	39	39 (two need to write)
2	44	39
3	40	40
4	45	40
5	38	41
6	43	40.40
7	39	40.92
		40.54

Given is that

$\alpha = 0.2$

$\therefore N = 10$

$\therefore (1-\alpha) = 0.8$

So, the formula becomes

$F_{t+1} = 0.2(A_t) + 0.8(F_t)$

i] For $A_1 = 39$, $A_t = F_t$ $\therefore F_1 = 39$

$F_{t+1} = 0.2 \times 39 + 0.8 \times 39 = 39$ $\therefore F_2 = 39$

ii] For $A_2 = 44$,

$F_{2+1} = 0.2 \times 44 + 0.8 \times 39 = 40$ $\therefore F_3 = 40$

iii] For $A_3 = 40$,

$F_{3+1} = 0.2 \times 40 + 0.8 \times 40 = 40$ $\therefore F_4 = 40$

iv] For $A_4 = 45$,

$F_{4+1} = 0.2 \times 45 + 0.8 \times 40 = 41$ $\therefore F_5 = 41$

v] For $A_5 = 38$,

$F_{5+1} = 0.2 \times 38 + 0.8 \times 41 = 40.40$ $\therefore F_6 = 40.40$

vi] For $A_6 = 43$,

$F_{6+1} = 0.2 \times 43 + 0.8 \times 40.40 = 40.92$ $\therefore F_7 = 40.92$

vii] For $A_7 = 39$,

$F_{7+1} = 0.2 \times 39 + 0.8 \times 40.92 = 40.54$

$\therefore F_8 = 40.54$

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Calculate mean squared errors →

Error	MSE
$A_t - F_t$	$\lvert A_t - F_t \rvert^2$
-	-
5	25
0	0
5	25
-3	9
2.60	6.76
-1.92	3.69
	$\Sigma = 2$

Take $n = 6$
Since we calculated F_t for 2 - 7 weeks

$$\therefore \text{MSE} = \frac{25 + 0 + 25 + 9 + 6.76 + 3.69}{6} = 11.58$$

* Disadvantages -
- More complex than SMA : More recursive calculations
- Can Overreact : Responds to short-term noise since α is high.

* Comparison bet^n Moving Averages →

	MA Type	Weights	Reacts to change	Complexity	Lag
1)	SMA	Equal	Slow	Easy	High
2)	WMA	Recent > older	Moderate	Medium	Medium
3)	EMA	Recent (1)	Fast	Higher	Low
4)	CMA	All past = Equal	very slow	Easy	Very High

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* Autoregressive (AR) Models →

- Autoregressive (AR) models are a class of statistical models used for time series analysis and forecasting.
- These models are based on the principle that the current value of a time series is a linear combination of its previous values and a noise term.
- An autoregressive (AR) model expresses the value of a time series at time t as a function of its past values. The idea is that the past values of a time series contain useful information about future values, which can be leveraged to make predictions.

The general form of an autoregressive model of order p is →

AR(p) →

$$AR(p) = X_t = \Phi_1 X_{t-1} + \Phi_2 X_{t-2} + \dots + \Phi_p X_{t-p} + \epsilon_t$$

where,

- X_t is the value of the time series at time t.
- $\Phi_1, \Phi_2, \dots, \Phi_p$ are the autoregressive coefficients that he estima
- p is the order of the model (lags)
- ϵ_t is the error term or noise

* key concepts →

1] Order of the AR model (p):

- The number p represents the number of lag values (previous time steps) used in the model.
- For e.g. AR(1) uses the previous time step's value, AR(2) uses the previous two, and so on.
- Choosing the optimal p is critical for model accuracy and can be done through techniques such as ACF & PACF.

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2] Stationarity :

- For an AR model to work effectively, the time series should be stationary.
- A stationary series has constant mean, variance and autocorrelation over time.
- Non-stationary series often require differencing or transformations to make them stationary.

3] Autocorrelation :

- Autoregressive models rely on the correlation of the current value with past values.
- The strength of the relationship between x_t and its past values is captured by the coefficients.

$$\phi_1, \phi_2, \phi_3, \ldots, \phi_p.$$

* Generalized Eqⁿ is →

$$AR(P) = C + \sum_{i=1}^{p} \phi_i X_{t-i} + \epsilon_t$$

e.g. → Let's consider for AR(1), where $p=1$ →

$$X_t = \phi_1 X_{t-1} + \epsilon_t$$

This model states that the value of the series at time t depends on the value of the series at time (t-1) & some error term (ϵ_t).

- If $\phi_1 = 0.5$, the model implies that the current value is half of the previous value, plus some noise.
- If $\phi_1 = 0.9$, the current value is more strongly correlated with the previous value.

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* Moving Average (MA) models →

- A moving average (MA) model is a time series model where the current value of the series is expressed as a linear combination of past noise (error) terms.
- Unlike the simple Moving Average, which is a smoothing technique, the MA model in time series analysis is a stochastic model used to model the randomness (noise) in a time series.
- For an MA model of order q, written as MA(q) →

$$MA(q) = X_t = \mu + \epsilon_t + d_0 u_t + d_1 u_{t-1} + d_2 u_{t-2} + \dots$$

Generalized eqⁿ →

$$MA(q) = X_t = \mu + d_0 u_t + \sum_{j=1}^{q} d_j u_{t-j}$$

where, →

u_t or X_t = value of the time series at time t

μ = mean of the series

ϵ_t = error at time t

$d_0, d_1 \dots d_n$ = parameters of MA model

q = order of the model

* q value is founded by ACF plot.

e.g. → For $q=2$,

$$MA(2) = \mu + \epsilon_t + d_0 u_t + d_1 u_{t-1} + d_2 u_{t-2}$$

* When to use?

- When ACF shows significant spikes at initial lags & then drops off.
- When the data seems to white noise with a short memory of shocks
- In combination with AR terms as part of an ARMA or ARIMA model

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* Autoregressive Moving Average (ARMA) model →

- ARMA stands for AutoRegressive Moving Average, and it is a widely used model in time series analysis to describe and forecast stationary time series data.

* ARMA = AR + MA
  - AR (AutoRegressive) : The current value depends linearly on its own previous values.
  - MA (Moving Average) : The current value depends linearly on past forecast errors.

So, ARMA combines both components to better model complex time-dependent structures.

- eqⁿ →

  ARMA(p,q) Model →

  p → order of the AR part (no. of past values)

  q → Order of the MA part (no. of past forecast errors)

Mathematical Form →

$$X_t = a + \sum_{i=1}^{p} \phi_i X_{t-i} + \sum_{j=0}^{q} d_j u_t + \epsilon_t$$

where,

- X_t = value at time t
- ϕ_i = AR coefficients
- θ_j = MA coefficients
- ϵ_t = error
- a = constant

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* When to use ARMA?

- The time series is stationary
- The are pattern in both the value and the errors.

e.g → Dataset → Sales over 8 Days

Day (t) → 1 2 3 4 5 6 7 8

Sales (X_t) → 20 22 21 23 24 22 25 26

Let's assume; $c = 1$, $\phi_1 = 0.5$ (AR), $\theta_1 = 0.4$ (MA)

$\varepsilon_1 = 0$

For,

ARMA(1,1) ⇒

$$X_t = c + \phi_1 X_{t-1} + \theta_1 u_{t-1} + \varepsilon_t$$

$\because \varepsilon_1 = 0$ $\quad \therefore u_1 = 0$

we will start from Day 2 →

i] For Day 2 → $\therefore t = 2$

$\hat{X}_2$ → Means Forecasted

$$\hat{X}_2 = c + \phi_1 \cdot X_1 + \theta_1 u_1 \cdot$$

$$= 1 + 0.5 \times 20 + 0.4 \times 0 \cdot$$

$$= 1 + 10$$

$$\therefore \hat{X}_2 = 11$$

$$\therefore u_2 = 22 - 11 = 11 \quad \text{(error)}$$

ii] For Day 3 → $\therefore t = 3$

$$\hat{X}_3 = 1 + 0.5 \times 22 + 0.4 \times 11 = 1 + 11 + 4.4 = 16.4$$

$$\therefore u_3 = 21 - 16.4 = 4.6 \quad \text{(error)}$$

iii] For Day 4 → $\therefore t = 4$

$$\hat{X}_4 = 1 + 0.5 \times 21 + 0.4 \times 4.6 = 1 + 10.5 + 1.84 = 13.34$$

$$\therefore u_4 = 23 - 13.34 = 9.66 \text{ (error)}$$

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iv) Day 5 → $t=5$

$$\hat{X}_5 = 1 + 0.5 \times 23 + 0.4 \times 8.66 = 1 + 11.5 + 3.464 = 16.364$$

$$u_5 = 24 - 16.364 = 7.636$$

v) Day 6 → $t=6$

$$\hat{X}_6 = 1 + 0.5 \times 24 + 0.4 \times 7.636 = 1 + 12 + 3.0544 = 16.0544$$

$$u_6 = 22 - 16.0544 = 5.9456$$

vi) Day 7 → $t=7$

$$\hat{X}_7 = 1 + 0.5 \times 22 + 0.4 \times 5.9456 = 1 + 11 + 2.3782 = 14.3782$$

$$u_7 = 25 - 14.3782 = 10.6218$$

vii) Day 8 → $t=8$

$$\hat{X}_8 = 1 + 0.5 \times 25 + 0.4 \times 10.6218 = 1 + 12.5 + 4.2487$$

$$\hat{X}_8 = 17.7487$$

$$u_8 = 26 - 17.7487 = 8.2513$$

Day	Actual (X_t)	Forecast ($\hat{X}_t$)	Error (u_t)	u_t^2
1	20	-	0	0
2	22	11	11	121
3	21	16.4	4.60	21.16
4	23	13.3400	9.660	93.31
5	24	16.3640	7.6360	58.31
6	22	16.0544	5.9456	35.33
7	25	14.3782	10.6218	112.83
8	26	17.7487	8.2513	68.43

$$MSE = \frac{\sum u_t^2}{7} = \frac{510.39}{7} = 72.86$$

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* ARIMA (Auto-regressive Integrated Moving Averages)

- ARIMA is a statistical model used for forecasting and analyzing time series data that may have trends, noise and seasonality.
- It combines:
  - AR (AutoRegressive) part: uses past values to predict the future.
  - I (Integrated) part: uses differencing to make the time series stationary.
  - MA (Moving Average) part: uses past forecast errors for predictions.

* I component (Integrated component) →

- This component makes the series stationary by differencing
- Differencing means substracting the previous value from the current value.
- First difference:

$$Y_t' = Y_t - Y_{t-1}$$

- If the data is still not stationary, a second difference is applied:

$$Y_t'' = Y_t' - Y_{t-1}'$$

  - The value of d is the no. of differences applied
  - It handles the trend in data.

* Notation: ARIMA (p, d, q)

- p → No. of AutoRegressive (AR) terms
- d → No. of Differences needed to make data stationary.
- q → No. of Moving Average (MA) terems

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✱ Concepts

1] Stationarity: A stationary time series has constant mean, variance and autocorrelation over time. ARIMA requires the series to be stationary.

2] Differencing (d): Used to make the series stationary by subtracting previous values:

$$Y_t' = Y_t - Y_{t-1}$$

(d=1)

3] AutoRegressive (AR):

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + \varepsilon_t$$

4] Moving Average (MA):

$$Y_t = \theta_1 u_{t-1} + \theta_2 u_{t-2} + \dots + \theta_q u_{t-q} + \varepsilon_t$$

✱ Steps to Build an ARIMA Model →

1] Visualize the Data
2] Check Stationarity (Plot)
3) Difference the data to achieve stationarity (if required)
4] Identify p, d, q using:
- ACF (AutoCorrelation Function) → to choose q
- PACF (Partial AutoCorrelation Function) → to choose p

5] Build ARIMA(p, d, q) Model
6] Train the Model
7] Make Forecasts
8] Evaluate using metrics like MAE, MSE, RMSE.

✱ When to use?

- Time series is non-stationary.
- When there's a clear trend but no seasonality.
- when you want a flexible, interpretable model.

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* ADF Test For Hypothesis to make data Stationary:

- The ADF Test checks whether your time series has a unit root, i.e. whether it is non-stationary.
- Hypotheses →
  1) Null Hypothesis (H_0): The data has a unit root. (Non-stationary)
  2) Alternative Hypothesis (H_1): The data has no unit root. (Stationary)

* P-Value in ADF Test →

The p-value tells you how likely it is that your time series is non-stationary.

→ P-value > 0.05 •
- Fail to reject the null hypothesis (H_0)
  ↓
  You need differencing (d=1 or more)

→ P-value ≤ 0.05 →
- Reject the null hypothesis (H_0)
  ↓
  No need differencing (d=0 in ARIMA)

e.g → Solve e.g like ARMA

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